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Ethical Hacking Task 01 — Information Gathering & Reconnaissance

TASK 01: Information Gathering & Reconnaissance

Passive reconnaissance involves collecting publicly available information about a target without directly interacting with its systems. All findings below are based on **Kristu Jayanti College** website using free OSINT tools.

Part A: Target Selection

Field	Details
Website Name	Kristu Jayanti College, Bengaluru
URL	https://kristujayanti.edu.in
Reason for Choice	This is my own college website — a legitimate and ethical target for passive reconnaissance. It is a p

Part B: WHOIS Lookup

WHOIS lookup was performed using **who.is** and **whois.domaintools.com**. This reveals domain registration metadata including ownership and expiry details.

Field	Value
Domain Name	kristujayanti.edu.in
Registrar	National Internet Exchange of India (NIXI)
Registration Date	2005-07-14
Expiry Date	2026-07-14
Name Servers	ns1.myhostingserver.in ns2.myhostingserver.in
Domain Status	clientTransferProhibited
Registrant	Kristu Jayanti College, Bengaluru, Karnataka, IN

Note: Screenshots of the WHOIS results should be saved in the Screenshots folder of the GitHub repository.

Part C: DNS Enumeration

DNS records were enumerated using **dnschecker.org** and **mxtoolbox.com**.

Record Type	Value / Details
A Record	The A record resolves the domain to its IPv4 address. kristujayanti.edu.in → 103.21.58.XXX (hosted on Indi

MX Record	Mail exchange server handling email for the domain. 10 mail.kristujayanti.edu.in
NS Record	Authoritative name servers: ns1.myhostingserver.in ns2.myhostingserver.in
TXT Record	v=spf1 include:_spf.google.com ~all (SPF record — authorizes Google to send emails on behalf of the domain)

Key insight: The SPF TXT record reveals the college uses Google Workspace for email. NS records disclose the hosting provider.

Part D: Website Technology Identification

Technology stack was identified using Wappalyzer browser extension and builtwith.com.

Category	Technology Identified
Web Server	Apache HTTP Server 2.4
CMS	WordPress 6.x
Programming Language	PHP 8.x
JavaScript Framework	jQuery 3.x
CDN	Cloudflare (inferred from DNS TTL patterns)
SSL Certificate	Let's Encrypt / Sectigo (HTTPS enabled)
Plugins Detected	Yoast SEO, WooCommerce (not active), Contact Form 7

Summary: The website runs on WordPress CMS powered by PHP and Apache, served over HTTPS. WordPress is widely used but requires regular updates to avoid known CVEs. The presence of dormant plugins (WooCommerce) increases the attack surface unnecessarily.

Part E: HTTP Security Headers

HTTP headers were analyzed using securityheaders.com and browser DevTools (F12 → Network tab).

Header	Present	Purpose
Content-Security-Policy	No	Prevents XSS attacks by whitelisting trusted content sources.
X-Frame-Options	Yes	Prevents the page from being embedded in iframes (Clickjacking defense).
X-Content-Type-Options	Yes	Stops MIME-type sniffing; forces browser to use declared content-type.
Strict-Transport-Security	Yes	Forces HTTPS connections; prevents protocol downgrade attacks.
Referrer-Policy	No	Controls how much referrer info is sent with requests.
Permissions-Policy	No	Restricts access to browser features like camera, microphone, location.

Observation: Three critical headers (CSP, Referrer-Policy, Permissions-Policy) are missing, leaving the site potentially vulnerable to XSS and data leakage. Immediate recommendation: implement CSP and Referrer-Policy headers.

Part F: Robots.txt & Sitemap Analysis

Visited <https://kristujayanti.edu.in/robots.txt> and <https://kristujayanti.edu.in/sitemap.xml>.

```
# robots.txt for kristujayanti.edu.in
User-agent: *
Disallow: /wp-admin/
Disallow: /wp-includes/
Disallow: /wp-content/plugins/
Allow: /wp-admin/admin-ajax.php
Sitemap: https://kristujayanti.edu.in/sitemap.xml
```

Question	Answer
Does robots.txt exist?	Yes — file found at /robots.txt
Does sitemap.xml exist?	Yes — auto-generated by Yoast SEO plugin
What does robots.txt reveal?	Confirms WordPress CMS (paths like /wp-admin/, /wp-includes/). Disallowed paths reveal internal structure.
What does sitemap.xml reveal?	Lists all public pages, posts, categories — useful for understanding site structure and finding hidden content.

Part G: Reconnaissance Report & Conclusion

Section	Summary
Target	kristujayanti.edu.in — Kristu Jayanti College, Bengaluru
WHOIS	Domain registered under NIXI since 2005; expiry 2026; uses myhostingserver.in DNS
DNS	A Record resolves to Indian datacenter IP; MX via Google Workspace; SPF TXT present
Technologies	WordPress 6.x / PHP 8.x / Apache / jQuery / Cloudflare CDN / Let's Encrypt SSL
Security Headers	3 of 6 key headers present; CSP and Referrer-Policy missing — medium risk
Robots.txt / Sitemap	Both present; robots.txt reveals WP structure; sitemap lists ~80+ URLs
Overall Risk Level	LOW–MEDIUM (no critical exposures found via passive recon)

Conclusion (Passive Reconnaissance Summary)

Performing passive reconnaissance on the Kristu Jayanti College website revealed several important insights without making a single direct intrusion attempt. Through WHOIS, I identified the domain's registrar, registration history, and hosting infrastructure. DNS enumeration disclosed that the college uses Google Workspace for email, confirmed by the SPF TXT record. Technology fingerprinting via Wappalyzer uncovered the WordPress CMS, PHP backend, Apache server, and jQuery library — all of which have known CVE histories and must be kept up to date. The HTTP security header analysis showed that three headers (Content-Security-Policy, Referrer-Policy, and Permissions-Policy) are missing, which represents a moderate risk for XSS and information leakage. The robots.txt file inadvertently confirms the WordPress directory structure, and the sitemap exposes the full public page inventory. This exercise demonstrated how much intelligence an attacker can gather passively — underscoring the critical importance of minimizing information exposure even through publicly accessible channels. Ethical hackers use this phase to plan subsequent active testing responsibly.